

Solutions

1. (10 pts) Consider the problem of finding the points on the ellipse $3x^2 + 2y^2 = 1$ at which the product xy is a maximum. Fill in the blanks below to set up this problem using the method of Lagrange multipliers.

(1.1) The function you need to maximize is $f(x, y) = xy$

(1.2) The system you need to solve using Lagrange multipliers is a system with $\frac{3}{\#}$ equations and $\frac{3}{\#}$ unknowns.

- (1.3) Using the method of Lagrange multipliers, write out the system of equations you need to solve for this problem. You **DO NOT** need to solve the system. Write your equations in the box below. (Do not just write the general equations.)

$$\begin{array}{ll} (1) & y = \lambda (6x) \\ (2) & x = \lambda (4y) \quad \text{OR} \quad \langle y, x \rangle = \lambda \langle 6x, 4y \rangle \\ (3) & \underline{3x^2 + 2y^2 - 1 = 0} \quad \text{OR} \quad 3x^2 + 2y^2 - 1 = 0 \end{array}$$

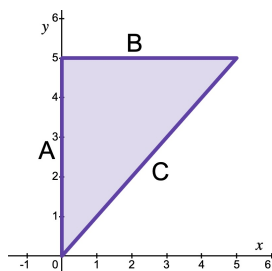
Note: $\nabla f = \lambda \nabla g \Rightarrow \begin{cases} f_x = \lambda g_x \\ f_y = \lambda g_y \end{cases}$ must follow from what was given for $f(x, y)$ in 1.1 and the constraint equation in 1.3.

2. (10 pts) For each true/false question, fill in the appropriate bubble. No work required.

- (2.1) ☐ (T) ☒ (F) If $\nabla f(a, b) = \langle 0, 0 \rangle$, then f must have a local maximum or minimum at (a, b) .
- (2.2) ☒ (T) ☐ (F) For a function $f(x, y)$ and point $P = (a, b)$: $f_x(a, b) = D_{\mathbf{u}}f(a, b)$ when $\mathbf{u} = \langle 1, 0 \rangle$.
- (2.3) ☒ (T) ☐ (F) The gradient $\nabla F(a, b, c)$ is perpendicular to the tangent plane to the level surface $F(x, y, z) = k$ at the point $P = (a, b, c)$.
- (2.4) ☐ (T) ☒ (F) If $f(x, y) = x \ln(x + y)$, then $(0, 0)$ is a critical point of f .
- (2.5) ☐ (T) ☒ (F) If a function $f(x, y)$ has continuous third-order partial derivatives then $f_{xyy} = f_{xxy}$.

3. (12 pts) Let $f(x, y) = x^2 + y^2 - 2x - 8y$ on the domain D such that $x \geq 0$, $0 \leq y \leq 5$, and $y \geq x$ (see figure below).

- (3.1) The maximum and minimum values of f on edge A have been provided. Find the maximum and minimum values of f on each of the remaining two edges of the triangle for the domain D . Complete the table with your final answers.



edge of triangle	A	B	C
max value of f	0	0	0
min value of f	-16	-16	$-\frac{25}{2}$

Show work below. You must provide justification for your answers for each remaining edge.

$$B: y=5, 0 \leq x \leq 5: f(x, 5) = x^2 + 25 - 2x - 40 = x^2 - 2x - 15$$

$$\text{Let } g(x) = x^2 - 2x - 15$$

$$g'(x) = 2x - 2 = 0 \Rightarrow x = 1$$

$$\underbrace{g(0)}_{=f(0,5)} = -15, \quad \underbrace{g(1)}_{=f(1,5)} = 1^2 - 2(1) - 15 = -16, \quad \underbrace{g(5)}_{=f(5,5)} = 5^2 - 2(5) - 15 = 0$$

$$C: y=x, 0 \leq x \leq 5: f(x, x) = x^2 + x^2 - 2x - 8x = 2x^2 - 10x$$

$$\text{Let } h(x) = 2x^2 - 10x$$

$$h'(x) = 4x - 10 = 0 \Rightarrow x = 5/2$$

$$\underbrace{h(0)}_{=f(0,0)} = 0, \quad \underbrace{h(5/2)}_{=f(5/2,5/2)} = 2(5/2)^2 - 10(5/2) = \frac{25}{2} - \frac{50}{2} = -\frac{25}{2}, \quad \underbrace{h(5)}_{=f(5,5)} = 2(5)^2 - 10(5) = 0 \text{ already found}$$

- (3.2) The only critical point of the function f is $(1, 4)$. Determine the absolute maximum and absolute minimum of value of f on the domain D . Put your answers in the spaces provided.

The absolute max value of f is 0 and the absolute min value of f is -17.

Show any additional work here:

$$\begin{aligned} f(1, 4) &= 1^2 + 4^2 - 2(1) - 8(4) \\ &= 1 + 16 - 2 - 32 \\ &= -17 \end{aligned}$$

4. (10 pts) Find the critical point(s) of $f(x, y) = 11x^2 + 2x^3 + 2y^2 + 4xy$. Write your final answer(s) on the line provided.

Critical point(s): $(0, 0), (-3, 3)$

Show work here:

$$\begin{cases} (1) f_x = 22x + 6x^2 + 4y = 0 \\ (2) f_y = 4y + 4x = 0 \Rightarrow 4y = -4x \Rightarrow y = -x \end{cases}$$

Substitute $y = -x$ into (1): $22x + 6x^2 - 4x = 0$

$$\Rightarrow 18x + 6x^2 = 0$$

$$\Rightarrow 6x(3 + x) = 0$$

$$\Rightarrow x = 0, x = -3$$

When $x = 0$ from (2) $y = 0$, so $(0, 0)$ is a critical pt

When $x = -3$ from (2) $y = 3$, so $(-3, 3)$ is a critical pt

5. (10 pts) Let $P = (-3, 6)$ be a critical point of a function $f(x, y)$, and let $Q = (0, 2)$ be a critical point of a function $g(x, y)$. Assume that $f_{xx}(P) = 4$, $f_{xy}(P) = 1$, $f_{yy}(P) = 0$ and $g_{xx}(Q) = -1$, $g_{xy}(Q) = 2$, $g_{yy}(Q) = -8$.
- (5.1) Apply the second derivative test to classify P . Fill in the bubble that corresponds to your answer. No credit will be given if more than one answer is selected.

- ☐ (A) f has a local maximum at P .
 ☐ (B) f has an absolute maximum at P .
- ☐ (C) f has a local minimum at P .
 ☐ (D) f has an absolute minimum at P .
- ☒ (E) f has a saddle point at P .
 ☐ (F) Second derivative test is inconclusive at P .

Provide a complete justification for your answer using the Second Derivative Test below:

$$D(P) = \begin{vmatrix} 4 & 1 \\ 1 & 0 \end{vmatrix} = 4(0) - 1^2 = -1 < 0$$

must include

- (5.2) Apply the second derivative test to classify Q . Fill in the bubble that corresponds to your answer. No credit will be given if more than one answer is selected.

- ☒ (A) g has a local maximum at Q .
 ☐ (B) g has an absolute maximum at Q .
- ☐ (C) g has a local minimum at Q .
 ☐ (D) g has an absolute minimum at Q .
- ☐ (E) g has a saddle point at Q .
 ☐ (F) Second derivative test is inconclusive at Q .

Provide a complete justification for your answer using the Second Derivative Test below:

$$D(Q) = \begin{vmatrix} -1 & 2 \\ 2 & -8 \end{vmatrix} = (-1)(-8) - 2^2 = 8 - 4 = 4 > 0$$

must include

and $g_{xx}(Q) = -1 < 0$ must include $\wedge$ or $g_{yy}(Q) = -8 < 0$

6. (9 pts) Let $f(x, y)$ be such that $\nabla f(x, y) = \langle y^4, 4xy^3 \rangle$, and let $x(s, t) = s^2t$ and $y(s, t) = t^2$. Apply the multivariable calculus chain rule to calculate $\frac{\partial f}{\partial t}$. Write only your final answer in the box provided. Your final answer must be simplified and written in terms of s and t .

$$\frac{\partial f}{\partial t} =$$

$$9s^2t^8$$

Show your work here:

$$\frac{\partial x}{\partial t} = s^2, \quad \frac{\partial y}{\partial t} = 2t$$

$$\frac{\partial f}{\partial t} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial t}$$

$$= y^4 (s^2) + 4xy^3 (2t)$$

$$= (t^2)^4 (s^2) + 4(s^2t) (t^2)^3 (2t)$$

$$= s^2t^8 + 8s^2t^8$$

7. (6 pts) Use implicit differentiation to find $\frac{\partial z}{\partial y}$ for $z^3 + zx + x^2y = x + 2yz$. Fill in **one** bubble that corresponds to your answer. Questions with multiple answers selected will receive no credit. No work required.

(A) $-3z^2 - x + 2y$

(B) $-\frac{3z^2 + x - 2y}{z + 2xy - 1}$

(C) $\frac{z + 2xy - 1}{3z^2 + x - 2y}$

(D) $-\frac{x^2}{3z^2 + x}$

(E) $-\frac{x^2 - 2z}{3z^2 + x - 2y}$

(F) $\frac{x^2 - 2z}{3z^2 + x - 2y}$

(G) $-\frac{3z^2 + x - 2y}{x^2 - 2z}$

(H) $-\frac{z + 2xy}{3z^2 + x}$

(I) $z + 2xy - 1$

(J) $3z^2 + x - 2y$

(K) $-\frac{z + 2xy - 1}{3z^2 + x - 2y}$

(L) $x^2 - 2z$

$$F(x, y, z) = z^3 + zx + x^2y - x - 2yz = 0$$

$$F_y = x^2 - 2z$$

$$F_z = 3z^2 + x - 2y$$

$$\frac{\partial z}{\partial y} = -\frac{F_y}{F_z} = -\frac{x^2 - 2z}{3z^2 + x - 2y}$$

8. (12 pts) Let $f(x, y, z) = z^2 + 2xy - x^2$, $\mathbf{v} = \langle -2, 6, 3 \rangle$, and $P = (1, 1, -3)$.

(8.1) a. The maximum rate of increase of f at P is $2\sqrt{10}$ or $\sqrt{40}$.

b. A unit vector that points in the direction of the maximum rate of increase f at P is

$\langle 0, 1/\sqrt{10}, -3/\sqrt{10} \rangle$ or $\langle 0, 2/\sqrt{40}, -6/\sqrt{40} \rangle$.

Show all work:

$$\nabla f = \langle 2y - 2x, 2x, 2z \rangle$$

$$\nabla f_P = \nabla f(1, 1, -3) = \langle 0, 2, -6 \rangle$$

$$\|\nabla f_P\| = \sqrt{0^2 + 2^2 + (-6)^2} = \sqrt{40} = 2\sqrt{10}$$

(8.2) Find the directional derivative of f in the direction of $\mathbf{v}$ at the point P . Fill in the bubble that corresponds to your answer. No credit will be given if more than one answer is selected.

(A) -6

(E) $\langle -2, 6, 3 \rangle$

(J) $\left\langle 0, \frac{12}{49}, -\frac{18}{49} \right\rangle$

(B) $-\frac{3}{\sqrt{10}}$

(F) $2\sqrt{10}$

(K) $\left\langle 0, \frac{12}{7}, -\frac{18}{7} \right\rangle$

(C) $-\frac{6}{49}$

(G) 7

(H) $\langle 0, 0, 0 \rangle$

(D) $\langle 0, 2, -6 \rangle$

(I) $\langle 1, 1, -3 \rangle$

☒ (L) $-\frac{6}{7}$

$$\vec{u} = \frac{\vec{v}}{\|\vec{v}\|} = \frac{\langle -2, 6, 3 \rangle}{\sqrt{4+36+9}} = \langle -2/7, 6/7, 3/7 \rangle$$

$$\begin{aligned} D_{\vec{u}} f(P) &= \nabla f_P \cdot \vec{u} = \langle 0, 2, -6 \rangle \cdot \langle -2/7, 6/7, 3/7 \rangle \\ &= 0 + \frac{12}{7} - \frac{18}{7} \\ &= -\frac{6}{7} \end{aligned}$$

9. (12 pts) Find the equation of the tangent plane to $f(x, y) = -\cos(xy)$ at the point $(\frac{2\pi}{3}, \frac{1}{4})$.

Answer: $z = -\frac{\sqrt{3}}{2} + \frac{1}{8}(x - \frac{2\pi}{3}) + \frac{\pi}{3}(y - \frac{1}{4})$

Show work here:

$$f(\frac{2\pi}{3}, \frac{1}{4}) = -\cos(\frac{2\pi}{3} \cdot \frac{1}{4}) = -\cos(\frac{\pi}{6}) = -\frac{\sqrt{3}}{2}$$

$$f_x = y \sin(xy) \Rightarrow f_x(\frac{2\pi}{3}, \frac{1}{4}) = \frac{1}{4} \sin(\pi/6) = \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}$$

$$f_y = x \sin(xy) \Rightarrow f_y(\frac{2\pi}{3}, \frac{1}{4}) = \frac{2\pi}{3} \sin(\pi/6) = \frac{2\pi}{3} \cdot \frac{1}{2} = \frac{\pi}{3}$$

10. (9 pts) Estimate $\Delta f = f(2.02, 0.96) - f(2, 1)$ assuming that $f(2, 1) = 8$, $f_x(2, 1) = 12$, $f_y(2, 1) = -32$. Simplify your answer.

Answer: $\Delta f \approx 1.52$

Show work here:

$$\begin{aligned} f(2.02, 0.96) &\approx L(2.02, 0.96) = 8 + 12(2.02 - 2) - 32(0.96 - 1) \\ &= 8 + 12(0.02) - 32(-0.04) \\ &= 8 + 0.24 + 1.28 \\ &= 9.52 \end{aligned}$$

$$\Delta f = f(2.02, 0.96) - f(2, 1) \approx L(2.02, 0.96) - f(2, 1) = 9.52 - 8 = 1.52$$

OR use

$$\Delta f = f(x, y) - f(a, b) \approx f_x(a, b) \Delta x + f_y(a, b) \Delta y$$